

THE MONTHLY REVIEW
OF
DENTAL SURGERY:
THE JOURNAL OF THE BRITISH DENTAL
ASSOCIATION

No. VIII. OCTOBER, 1880. Vol. I.

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VOL. I.

The subjects rightly embraced in a Medical education, and the degree and manner in which those subjects should be respectively studied, have been freely discussed in many places during the last few weeks. Dr. Michael Foster in an “Address in Physiology,”^[1] of unsurpassed interest, contends without contradiction that no medical subject—now that the entrance upon medical studies is preceded by a tested preliminary education—need be studied as heretofore as a mere mental training, and proposes that topographical anatomy, which has hitherto been so studied, should, to a certain extent, give way in favour of a more complete knowledge of physiology. The address must, and no doubt will, be read by all interested in medical education, whether general or special. The following quotation will answer our present purpose:—

¹. Address in Physiology, delivered at the Annual Meeting of the British Medical Association, 1880.—Published in *British Medical Journal*, August 21st, 1880.

“I think I am not overstating the case when I say that, in the two years (or less than two years) which the medical student devotes to studies other than clinical, 60 or 70 per cent. of his time—in some cases even more—is spent on the study of topographical anatomy. That study may be regarded in two lights—as a discipline, and as practical useful knowledge. The late Dr. Parkes, in a remarkable introductory address which he delivered at University College, London, many years ago, insisted most strongly that its value as a discipline was far higher and more precious than its direct utility; and I imagine that the more one reflects on the matter, the more clearly this will appear. The details of topographical anatomy have this peculiar feature, that, though they can only be learnt with infinite pains and labour, unlike other things hard to learn, they vanish and flee away with the greatest ease. I would confidently appeal to my audience of practical men, how much of the huge mass of minute facts, which in their youth they gathered with so much toil, remained fresh in their minds two years after they passed the portals of the College; and how much now remains to them beyond a general view of the parts of the human frame, and a somewhat more special knowledge of particular regions, their acquaintance with which has been maintained by more or less frequent operations. I would confidently ask them what is the ratio, in terms of money or any other value, which the time spent in those early anatomical struggles—say over the details of the forearm—bears to the amount of that knowledge remaining after twenty, or ten, or even five years of active practice, or to the actual use to which that knowledge has been put.

“No, it is as a discipline, and not for its practical utility, that anatomy has been so useful; and this, indeed, may frequently be recognised in the questions set at examinations. When the candidate is expected to describe, within the error of a few *millimètres*, the structures traversed by a bayonet thrust obliquely through the neck, or is invited to reproduce written photographs no less exact of the parts which, from skin to skin, underlie a triangle or quadrangle drawn in ink on the front or back of the thigh, it is clear that the examiner has in view, not the needs of practical life, but an easy means of testing the proficiency of the student in mnemonic gymnastics. Of the value of anatomy as a discipline, there can be no doubt. In past years, it has served as the chief culture of the medical student—as the chief means by which the rough material coming up to our great medical schools were trained to habits of accuracy, of exactness, of patient careful observation; and their memories

strengthened by exercise for the subsequent strain which would have to be put upon them by more strictly professional learning. In this aspect, the very sterility of the subject was a virtue. The mere fact that the separate details seemed to hang loosely, isolated in mental space, held together by no theory, by no ideas, inasmuch as it made the learning a harder task, increased its disciplinary value. Most wisely did the leaders of our profession insist that no trouble or expense should be spared to afford the neophyte this preparatory scientific training; and that, as far as examinations and the like can go, no pains should be spared to compel him to avail himself of the opportunities offered. Indeed, viewed as a branch of education, the machinery of anatomical instruction has for many years past not been equalled by any."

Professor Burdon-Sanderson in his introductory lecture, says,—

"The precious years which immediately precede a man's entry on professional duty, are far too valuable to be wasted in learning anything he does not intend to retain."—*British Medical Journal*, October 9th, 1880.

The observations of these most distinguishing physiologists and teachers, support the view which has been taken respecting the education of the Dental Surgeon, from the time the College of Surgeons was in 1857 asked to establish a Dental department, up to the determination of the Dental curriculum by the Medical Council in 1879.

An education equal in degree, but different in kind to that of the General Surgeon—an education which embraces a knowledge of the general principles of Medicine and Surgery, and a special, precise, and practical knowledge of Dental Surgery, was asked and given, the degree of education progressing as the attendant circumstances allowed, up to its present advanced condition.

The place assigned by Dr. Michael Foster to topographical Anatomy in Medical education, will certainly hold good in the education of the Dental Surgeon. Physiology and Chemistry, subjects now inseparable and of surpassing interest, are equally necessary to the Special and General Surgeon.

The position of Medicine and Surgery is not quite parallel. The general principles of each must be thoroughly known, but it is not necessary that the Dental Surgeon should be practically acquainted with all the details that pertain to any other branch of either than that which he selects to practice. It is not needful that he should become a skilful midwife or oculist, or that he should be skilled in the treatment in any other class of diseases than those to the treatment of which he proposes to devote his life. For if he does acquire such detailed knowledge, it will, when no longer used, "fade like a raindrop

upon a porous stone,” and the time devoted to the acquisition will have been wasted, and, perhaps, worse even than wasted, for the subject may have been studied at the cost of neglect of the practical study of his speciality. It does not admit of doubt that the high degree of manual skill, without which the professed dentist is but a shameless pretender, can be acquired best, perhaps can be fully acquired only during youth; that if the acquisition be put off till a medical education is completed, the golden opportunity will have been lost. Mr. Fawcett tells us that the blind may acquire manual skill sufficient to secure independence, but that it can be gained only during youth. The adult blind, he says, have a greatly diminished power of learning.

The time may come when the principles of Medicine and Surgery will be taught before their special application to any particular class of disease, and their modification resulting therefrom is entered upon. Till that time arrive, it will be in the interest of the Dental Surgeon to study with the utmost care the general principles and their application to his own speciality, and to acquire, while he can, a very full and perfect knowledge, practical, as well as scientific, of Dental Surgery, before he enters upon a detailed study of their application to any other branch or subject of Medicine or Surgery, the practice of which he does not propose to follow, and a trustworthy knowledge of which can not be retained or extended in the absence of continuous general practice.

The qualified surgeon who has devoted himself to the practice of dentistry, may be legally qualified to treat any and every kind of case, but would he be morally right in undertaking the treatment of a case, say of fever or of stone? all his knowledge of which diseases lies in a misty memory of facts brought before his notice, and studied for a pass examination in long past student days. As a matter of fact, the dental surgeon of necessity limits his practice to his speciality, and the general surgeon as a further fact, where selection is possible, declines to take Dental cases, though legally entitled so to do. In the absence of special training, he cannot be expected to possess the special knowledge requisite to successful treatment, and to add the Dental to the over loaded Medical curriculum, would be to greatly increase the rejections which, in the case of the College of Surgeons, have already risen to the formidable amount of upwards of thirty per cent. of the candidates for diplomas. Hence it is that Dr. Michael Foster, in asking for more physiology, asks for less topographical anatomy.

The Dental curriculum requires for its honest fulfilment, the whole of the assigned four years, and more must not be attempted in the same four years, for to repeat the words of Professor Burdon-Sanderson,—“The precious years which immediately precede a man’s entry on professional duty, are far too valuable to be wasted in learning anything he does not intend to retain,” to which may be added, or which his subsequent occupation will not allow him to retain. If then more medical knowledge is required of the student than is embraced in the dental curriculum, more time must be given for its acquisition.

Transplantation and Replantation.

(Continued from page 425.)

At the period when transplantation was systematically performed, the operators kept themselves supplied with an assortment of dried teeth, *i.e.*, sound teeth, obtained generally from the dead subject, to make use of in case the tooth to be transplanted should not prove adaptable to its new abode. The process is not spoken favourably of, as the results were mostly unsatisfactory, but at times such teeth became perfectly firm, and even resisted the greatest efforts at their removal. We can hardly for one moment suppose that teeth in the condition these were could have ever become attached to a living alveolo-dental membrane, and the difficulty of accounting for this firmness would have been great indeed, had it not been already solved for us. In Langenbeck's *Archiv. für Chirurgie*, vol. iv., is a paper on "The Replantation and Transplantation of Teeth," by Dr. A. Mitscherlich, which deserves to be better known, and which, apparently, cannot have come under the observation of those who recently contributed to the *Lancet* on the subject, the excuse for which is less, since it has been translated into English.^[2] The author, in addition to much interesting and original matter, records the following experiment. In the upper jaw of a dog of a year old, Dr. Mitscherlich inserted into the socket of an incisor he had removed, a similar tooth taken from a dog's skull, and which he retained *in situ* by means of a silver wire passed through a hole in the tooth, and a hole bored through the alveolar process of the jaw. "After six weeks the dog was killed, having been given during the last few days three grains of picronitrate of potash three times a day; the carotids were immediately injected. The muscles, like the gums, were coloured yellow: neither, however, in the implanted tooth nor in the sound ones was any alteration of

colour perceptible. The silver wire was porous, and no longer held the tooth; it was removed. The tooth was quite firmly seated, and could not be moved in the least by the fingers. The gums, as in the remaining teeth, were accurately applied both to the alveolar process and also to the tooth itself, and nowhere could any alteration be found in it. The tooth was sawn through lengthways, together with the upper jaw, with a fine saw, so that the pulp cavity was laid bare in its whole extent. The latter was only filled with a little detritus, and no trace of the pulp was discoverable; none of the injection, too, had been forced into the cavity. The tooth was everywhere most intimately connected with the surrounding parts, and suppuration showed itself nowhere. Of the periosteum, on the other hand, there were only in a few places some small remains discoverable. On the posterior surface of the tooth two small cavities were visible; the larger of which lay more towards the point of the fang, and extended to the pulp cavity; they were filled with a soft substance, and their walls appeared roundish. These appearances were entirely confirmed by microscopical examination, inasmuch as only in a few places, especially on the anterior surface of the tooth, were traces of the periosteum to be demonstrated; where this was absent the tooth was eaten away, and its absorption had proceeded in such a manner, that a multitude of globular elements appeared on the section, resembling the fragments which we find in ivory pegs which have been bored into bones, and retained in them a considerable time; in the two above-mentioned cavities the absorption had proceeded farther and farther, and had at last attained its above-mentioned great extent. The cementum could still be demonstrated in certain places, it was, however, absorbed in the greatest part of its extent. In the cavities of the tooth substance, masses of bone were embedded, they were applied to the walls of the cavities without any kind of intermediate substance, and so held the tooth with such extraordinary firmness. This osseous deposit, which was directly connected with the alveolar processes, was freely traversed with blood vessels, which sometimes extended themselves close up to the tooth substance. It was also so fully developed, that the process had to be looked upon as fully accomplished, and therefore, a later exfoliation of the tooth was not to be expected. The dentine itself showed nothing abnormal.”

[2.](#) “Archives of Dentistry,” Edited by Edwin Truman, vol. i., p. 169.

We have quoted the author of the foregoing at some length, as some of his conclusions upon this interesting experiment might seem antagonistic to our own. Thus, whilst we agree with him in concluding that a dead tooth becomes united to a living jaw by a certain amount of absorption of its fangs by osteoblasts, and a subsequent calcification of such tissue, we do not believe such process as fully accomplished or permanent. The valuable researches of Tomes and De Morgan show that during life a continual process of formation and absorption is taking place in bone, whilst the former has also pointed out that the process of absorption in the fangs of temporary teeth is one alternating between absorption and deposition, though in the end in favour of the former, *i.e.*, that the osteoblasts which effect the absorption of the dentine often become calcified, but are again eventually decalcified and become active osteoclasts. What determines these bodies to act, so to speak, in a positive or a negative capacity would be most interesting to discover, and, in considering the subject, we can hardly overlook the views of Mr. Bridgman, who compares the process of bone formation and decalcification to what occurs in an electrolytic cell, where, by changing the direction of the current, the electrodes assume precisely opposite functions.

If, then, we could discover the conditions that would preserve the bone tissue, in connection with the dentine, from undergoing decalcification, we might, with every prospect of success, transplant dead teeth, but which, as at present carried out, invariably, we believe, eventually, suffer the fate of ivory pegs introduced into the extremities of bones in disunited fractures. Indeed, this result, although more slowly effected, appears nearly always to follow in those cases where the transplanted or replanted tooth has lost much of its alveolo-dental membrane; whether the excising of a portion of the fang, as advocated by Magitôt, will prove beneficial or otherwise, remains to be seen.

Porcelain teeth having indentations in their fangs have been suggested, first, we believe, by Mitscherlich,^[3] and again by a recent writer in the *Lancet*. The former actually attempted two cases, and with the success we should have anticipated, for when after four or five weeks the caoutchouc splint which retained them *in situ* was removed they immediately followed the splint and fell out. "Ossification had not taken place; the entirely heterogeneous mass of stone had acted as a foreign body, produced granulation and suppuration, and so prevented union."

[3.](#) *Op. Cit.*

In conclusion, we think enough has been stated to show that transplantation or replantation of living teeth, or at all events, of teeth having living alveolo-dental membrane may be exceptionally carried out with benefit to the patient; if the chances of permanent success be not very great the chances of injury are, we believe, small, and have been greatly exaggerated. No such case has come under our observation, but then in all we have witnessed, the transplanted or replanted tooth has never been ligatured or otherwise forcibly retained in its alveolus. We believe many of the cases of failure, as probably those also of bone exfoliation, arose from this procedure. A tooth, after either operation, although at the time perfectly adjusted to its proper position, becomes, after a day or two, elongated from its socket and less firm; the result of effusion into the alveolo-dental membrane and about the tooth; as this material becomes organized it forms, no doubt, the medium of union between the dental and alveolar portions of that membrane. At all events, after a week or so the tooth again recedes into its socket, and as it does so becomes firmer and less sensitive to pressure; if our view be correct, the employment of a ligature or forcible retention of the tooth can only be objectionable. A very different matter, however, will be the adjustment of a plate contrived so as to protect the transplanted or replanted tooth from violence or pressure until its attachment is ensured.

Mercurial Amalgams.

BY M. G. CUNNINGHAM.

After twenty-five years of stubborn fight supporters of gold as a filling for decayed teeth accept the possibility of plastic material being in certain cases its superior; throughout this period I have been content to hold my peace and act entirely on my own judgment in the selection of material, as however, it seems to be the fact that a man who uses plastic filling without danger of being termed a “quack,” may speak, I would, through your kind agency, convey to brother Dentists my method of preparing metallic amalgams, which has saved me much trouble and my patients a large number of teeth.

In using amalgam, the first thing we ought to take into consideration is whether that which we are using and calling by that name is such, and I venture to say that in a very large number of cases it is no amalgam at all, but a concrete admixture of solid metals with liquid mercury. In the early days, when metallic precipitate of silver was employed, perfect amalgamation was not difficult to obtain, provided the precipitate had been in the first instance properly washed and carefully stoppered, the minute sub-division of the metal and absence of oxidation aiding largely to this result. The fillings of the present day are of a totally different character, coarse in grain, and of a nature to oxidize on even momentary contact with air, they become difficult to amalgamate with mercury, which, in itself, is a highly oxidizable metal, so that recourse is often had to a glass tube and violent agitation to produce that which is at best only a semblance of what it should be—a thoroughly homogeneous mass, that upon setting will retain a uniform texture and density proportionate to the constituents of which the fillings are composed. If, however, to the fillings and mercury be added a drop or less of sulphuric acid, either in the palm of the hand or mortar, it will be found that the metals will almost instantaneously amalgamate, whilst the oxides combining with the acid leave a residuum which, by its quantity, clearly shows what a very imperfect body could have been a so-called amalgam containing only a small portion of them. Washing in pure water at once removes all trace of acid, and a thoroughly reliable stopping can be at once produced from materials otherwise worse than useless.

Introductory Lecture to Course on Dental Mechanics.

BY DR. J. WALKER.

GENTLEMEN,—The authorities of the Dental Hospital of London have entrusted to me the heavy responsibility of lecturing on Dental Mechanics for this present Session of 1880 and 1881. I have undertaken this post with many misgivings; the subject is so wide and comprehensive, the study so important to you now, and the effect of good or bad teaching will so deeply affect your whole life, that I might well have hesitated before finally accepting the duties that will now devolve upon me.

I have taken a great interest in this Hospital and School from its early foundation, when the pupils were few, and the School and Hospital had, so to speak, to win its spurs; but in passing, I may here remark that that small class of students, then a new feature in London life, by their diligence, learning and conduct, have established throughout the country a name and reputation that you will have to perpetuate. It was by the combined efforts of the whole profession, and by the fact that its senior members were able to point back to so many successful students of this School and Hospital, now earnest practitioners in nearly every large town of the British Isles, that the leaders of our department of Surgery were enabled to bring this special branch before the notice of Parliament, and obtain a Bill which now governs the method and extent of the classes and hospital practice of all our schools.

As your lecturer, I may perhaps without egotism mention that I was one of the six gentlemen appointed as the first Assistant Dental Surgeons to this Hospital, while it was struggling through its first year of active life. I

remained at my post nearly ten years, until driven by increasing practice to relinquish the work, then, as now, carried on in your Hospital. It is the knowledge and experience I then gained, and the lessons I have since learnt as the result of treatment in private practice, that I have now to offer you—many failures and some successes, many abortive schemes, and some inventions that have stood the test of time.

My best thanks are due to the Managing Committee of this Hospital, for electing me to be the colleague of such men as Alfred Coleman, C. S. Tomes, and D. Lewis.

Gentlemen, I am fully conscious of the honour, and will do my best, if health and strength are given me, to redeem in part the loss you have sustained in the resignation of your late teacher, Mr. James Smith Turner. I know that you and your companions in hospital practice held him in high esteem. He has been a true friend to the London Dental School. He had become a ripe and experienced teacher, one fully alive to the best manner of treating his subject, passing by what the student might gain for himself from books, to dwell upon those details that a man of his experience could so fully grasp. But much as we miss him, we may yet congratulate ourselves that we have not to mourn over the death of so good and faithful a friend. He is now as actively at work in another sphere of Dental life; he is even working harder than ever in perfecting the work that has so prospered in the hands of Messrs. Tomes and Turner. He is consolidating the Dental Act of 1878, making every effort to render the Register of 1881 as perfect as may be, and to raise the standard of Dental Education throughout Great Britain. That the number of Dental Schools may be equal to what is now demanded by students, the schools be sound in teaching power, the men elected as teachers be conscientious in their newly appointed work, that full and complete courses on each Dental subject be delivered—these are a few items of his daily work. The end and object of his endeavours is that the students in the various schools may gather wider stores of knowledge during their curriculum, so that the various examining bodies may see their way to enforce a higher standard of examination than is now enforced to obtain the L.D.S. Diploma, and the coveted power of registration.

Those gentlemen who were members of Mr. Turner's class last year will join with me in wishing him long life and energy to complete his self-imposed task.

To come to the special subject of my lecture. At the risk of provoking the well-known retort of “nothing like leather,” I venture to assert that no man can ever prove himself a good Dental Surgeon unless he is a skilled artist in Dental Mechanics. To kindle a spark of my own enthusiasm for the subject, to fan that spark into a flame that shall burn brighter and brighter in your life until you lay down the file and the engraver, with a sense that you have done some good work in the world, is at once my endeavour and my duty. A painter or sculptor of eminence in his profession is frequently the leader in the fashionable world, he is feasted by the city guilds, his company is sought by the rich and noble, he has the *entrée* to the literary circles of every capital, yet the work of the artist is, at best, but to reproduce a faint imitation of nature in cold marble or on inanimate canvass. The subjects of your handiwork will be full of life and animation.

What is the necessary training for such accomplishments? A full and perfect realization of all the forms of human beauty, and of woman’s beauty in particular. Why do I dwell so much on the complete form of beauty? Because no face can be perfect in beauty, unless its features each and all are in harmony. The teeth have a peculiarly marked position in relation to the features: one missing link in the circle will attract attention and mar the harmony of an otherwise lovely face, like a false chord in music.

Projecting, discoloured, irregular, misshapen, crowded teeth, all tend to destroy symmetry. If so in the natural course of dentition, how much more in the artificial!

I would urge you all to undertake at starting a thorough study of the normal bones of the skull.

Normal bones of the face.—In the anatomical class at your general hospital you will study the bones of the face in considerable detail, but there your attention will be directed to the common or general characters of the bones. You will there have to learn the usual shape of the bones, their processes, ridges, grooves and depressions; you will be shewn the characters, not only by which you may at once recognise them, but which you may always recognise in them. I cannot too strongly urge you to master all these details.

But, Gentlemen, here we have to study these bones in a practical manner; we have to look upon them as parts of the living countenances of our patients, and as no two faces are exactly alike, a study of individual faces is

necessary, as a groundwork for your success in practical mechanical dentistry, and you must study individual specimens of each of the facial bones. A careless observer of a crowd of negroes might think they were all alike, because each had a black skin, woolly hair, retreating foreheads, thick lips and white teeth. But yet a close observation would quickly tell him in truth there came behind all these coarse resemblances, minute, but noteworthy differences, differences which he would be compelled to take note of before intercourse with them would be possible. In the same way a general anatomist merely points out to you how all palates are alike, and I want to go farther and shew you with equal truth how no two are alike, but all differ. Depend upon it, gentlemen, your usefulness and success will vary with your skill in perceiving these lesser differences which characterise individuals. The best name I can give to this study is comparative human anatomy.

Let us take some examples of what I mean. The upper jaw bone is the most complex of all the bones of the face. Looking at its central part or body we are first of all struck with the cavity in it—the antrum of Highmore. Now, if you take a hundred bones, you will not be able to find two antra exactly alike, but they will differ in size, in shape, in depth, in width, and in size of their angles and inclination of each of their walls. And all these peculiarities influence the countenance, and must, therefore, be studied before you can hope to be successful in replacing the lost dental organs. The high cheek bone of the Scotchman is a very familiar example of the effect of a variation in the antrum.

The alveolar process.—The natural setting of the teeth varies also in its depth, thickness, smoothness, irregularity, and most importantly in its curve, which may be a broad, open semi-circle, or a narrow semi-ellipse. The nasal process, too, varies as much in different specimens; you will find differences in length, breadth, in the angle it forms with the body of the bone, and with the frontal bones; all these particulars modify the shape of the nose, and as I shall have to point out to you, no feature is more worthy of your careful study than the human nose in its numberless varieties. The malar process of this bone has similar varieties. Notice again the palate plate how it differs in breadth and arch, and so modifies importantly the roof of the mouth, to which a denture has been adapted. The malar bones are unlike in thickness, the size of their angles, length of their offspring processes, and in the exact mode of articulation with neighbouring bones.

See too, how frontal bones vary, in one case a broad, bold line forehead, in another overhanging, in a third narrow and pointed, and you meet with infinite varieties between these extremes.

In passing to the nasal bones, not only must we notice how they differ in length and breadth, and the level of their edges, but that the shape of their arch is constantly varying; it may be broad and rounded, or narrow and high, even to sharpness. This depends upon the prominence forward of the bony nasal septum, the interval between the nasal processes of the upper maxilla, that is to be bridged over the breadth of the nasal bones, and the exact mode of their articulation with the upper jaw bone. Not alone does the usual arch differ thus, but most obviously on the angle it forms with the frontal lines.

From your own observation you will at once grant me that noses vary as much as families; in fact I am inclined to think that there is a good deal to be said for Mr. Shandy's philosophy of noses. The cartilages of the nose play a most important part in the shape of the organ, and demand your study as much as the bones. Each variation in the shape of the nose has a corresponding variety of upper lip, and the correlations between these two must be most carefully attended to. Granted that these differences are so numerous, you must admit that the nose must have primary importance in the estimate of the Dental Surgeon, when called upon to restore the lost Dental organs. I may remark that although I am examining noses every day of my life, I have never yet found one assuming a direct line with the other central lines of the head and face.

To arrive at a just appreciation of the effect of these bones on the lines of the face you must examine them in the articulated skeleton, not in one instance but in many—fifty or even a hundred—make weekly visits to the Museum of the Royal College of Surgeons, and there examine all the specimens of articulated skulls and skeletons, until you fully grasp the meaning of comparative human anatomy—the size, the shape, the relative acuteness of angles, the proportions of the different parts. It is this relationship, the articulation of each bone with the other bones of the skull that is of primary importance to the Dental Surgeon.

The last bone that I shall mention to you this evening is the lower jaw, perhaps the most important of all. You will all soon be taught that it has a body, a symphysis, a ramus with its condyle, coronoid process and sigmoid

notch, an alveolar process, and various tubercles, ridges, spines, grooves, and depressions. But beyond all such facts, be at pains to notice, gentlemen, how all these various parts differ in different specimens. The changes in the angle of the bone that are met with at different ages are notorious, but you will have to learn that the angle of every adult differs, that each form of countenance has its special maxillary angle, nor are the depth, thickness, curve, obliquity and relative prominence of different parts of the bone one whit more constant, and if you would succeed in fitting artificial dentures to a lower jaw, these individual peculiarities of the bone must be carefully studied.

The Dental organs will be presented in full detail by my colleague, Mr. C. S. Tomes, but I should fail in the one point of my brief sketch, if I did not refer you to the fact that no circle is found exactly corresponding with a second in the articulation of the thirty-two teeth implanted in the maxillary bones. My remarks culminate in this apparently strange contradiction, no two sets of teeth ever describe the same circle at any age. The differences in children are only slight, yet a difference exists; the older the subject, the greater the contrast visible. Yet harmony exists in the lines of the face; once acknowledge this and you will perceive the labour and investigation necessary to make you grasp the subject in its broadest sense.

Take, for instance, a patient at the age of sixty, with edentulous jaws, requesting artificial dentures to be prepared at your hands. For such mechanism to be successful in the restoration of the contour of the face, it will be necessary that it shall harmonise with the features; you must carry your perception backwards to the appearance that this face presented when he had only attained the age of thirty: this will be the art and science expected at your hands.

Not to lengthen this my introductory lecture, by labouring to define the various types of English faces, when the bones are covered with the soft parts, I have selected a few outlines of faces such as a sculptor and painter would study. Cast your eye upward, you will see that each possesses its own characteristic and alterable features. So in life; and if you gentlemen are to be true Dental Surgeons, you must rise to the ideal of artistic mechanics. If I am to benefit you to the full bent of my wishes, you must study nature in all its variety—nature when presented to you as destroyed by premature disease and death, to be restored by your hand to its original conformation.

Remember that use must follow beauty, one cannot be dissevered from the other if you would obtain results, complete in power of mastication, speech, durability, and appearance.

To win success in appearance, you must study the irregularities of the natural projection of circle, and the character of circle. If any irregularities are observed, take a model cast of the mouth, so that you can compare the natural organs whilst the artificial are in progress of arrangement; modify these irregularities, but do not efface their existence altogether. The general conformation of the Dental organs and the face must be your special study. The colour of the teeth to be selected must receive at your hands great care: compare the various tints at your disposal, ascertain if a perfect self colour, or a tinted, shaded, or stained tooth is the most pleasing; which will harmonise best with the complexion, producing a natural effect. It will be your object to disguise that the new introductions are foreign bodies, endeavour to make them appear as if they possessed life. Many colours absorb so much light that at night the appearance they present is black and death-like. Avoid such shades, select those that reflect light; in many cases the teeth should be almost transparent. The colour of the hair, the nature of the complexion will guide you in this endeavour; hair and complexion must harmonise. A heavy, wavy head of hair, dark and massive, with bronzed face would indicate a strong shaded tooth, solid in character and non-transparent, yet a colour that will reflect rather than absorb the rays of light. A Saxon face with fair hair will strongly puzzle you at times; the pearl hued, thin and transparent teeth, as a rule, are appropriate for such a face.

When articulating the dentures, every care must be taken and much thought and study bestowed to adjust the depth and height of the superior and inferior dentures to the length and depth, to the thickness or thinness of the lips. Note and estimate the loss of structure by absorption, and supply in proportion to the loss.

The circle and projection as above alluded to will be your special study. Give hours to produce a natural expression. Propose to the patient a short walk in your operating room; if a good corridor is at your disposal, so much the better, use it. By so doing you will ascertain what, if any, old habit of contortion of features is indulged in by the patient; the character of the laugh, if the lips are raised, and how much; at times only the tips of the teeth are manifest, at others the full lip is raised exposing to view the crown

of the tooth and the alveolus, even to the lower margin of the meatus of the nose. Arrangements equal to all these expressions must be adopted. The how, will be told you in future lectures.

Then, again, the smile; the smile of a Desdemona and the smile of an Iago! yet how much in a smile! Harmonise your mechanism so that the smile of the patient shall be rendered as natural and full of meaning as art can make it. The movements of the lips and tongue in speech must be as carefully studied; the eloquence of the orator and the no less eloquent prattle of a pretty woman. During the visit of your patient strike out a conversation that shall put him at his ease, and show him at his best; this will aid you in observing what is required to render conversation easy to himself and acceptable to his hearers. Notice that the head is never carried in a perpendicular line with the body; at times it leans to the right or the left, the effect of this inclination is to lengthen in appearance the teeth of the incline. Notice also, many patients have the ungainly habit of twisting the lips out of the natural line of the head and face.

It is by attention to all these details that the results of your work will be noble, your claims to reward great, your satisfaction not less in its kind than that of the sculptor or the painter; whilst they can only charm the happy, it is yours to relieve the suffering, to alleviate pain, and even to prolong life and restore lost beauty.

Special General Meeting of the Midland Counties Branch of the British Dental Association.

A Special General Meeting of the Midland Counties Branch of the British Dental Association, was held on Wednesday, 6th October, in the Memorial Hall, Albert Square, Manchester. The President of the branch, H. Campion, Esq. (Manchester), presided, and there were also present, Messrs. S. Wormald (Treasurer), Stockport; Dr. W. H. Waite (Secretary), Liverpool; W. H. Nicol (Leeds); R. E. Stewart (Liverpool); Dr. D. A. Wormald (Bury); T. Murphy (Bolton); Q. Renshaw (Rochdale); W. H. Ridge (Stafford); T. Mahonie (Sheffield); B. Harding (Manchester); T. C. Parson (Clifton); H. Marsh (Manchester); T. Dilcock (Liverpool); D. Dopson (Liverpool); L. Matheson (Manchester); W. Dykes (Manchester); W. Headridge (Manchester); J. G. Roberts (Liverpool); W. Taylor (Batley); W. Shillinglaw (Birkenhead); J. S. Crapper (Hanley); and R. Rogers (Cheltenham).

THE MINUTES OF THE LAST MEETING.

The first business before the meeting was the reading of the minutes of the last meeting. They were taken as read.

The following letter was read by the Secretary (Dr. W. H. Waite):—

“To the Hon. Sec. of the Midland Branch of the British Dental Association.

“DEAR SIR,—I have much pleasure in informing you, that at a meeting of the business Committee of the representative Board of the British Dental Association, held on the 1st instant, Edward Saunders, Esq., in the chair, the proposed bye-laws having been approved of, the Midland Counties Branch of the British Dental Association was formally affiliated with the Central Body.

“According to the desire of the meeting, I have to convey to you our best wishes for the success of your efforts.

“Yours very truly,
“JAMES SMITH TURNER,
“*Hon. Sec., British Dental Association.*”

The report which was then read by the Secretary, was as follows:—

REPORT OF COUNCIL.

As this is a Special, and not an Ordinary General Meeting, and as the Council have only met once, there is not much upon which they can report. In accordance with the resolution adopted at the last meeting, circulars inviting practitioners in the Midland District to join the Association, were issued to every name on the Register. Up to the first of August, the Secretary had enrolled 33 members and 6 associates, and since that date there have been six applications for membership, five of which the Council have this day admitted, viz., Messrs. R. Rogers, Cheltenham; T. C. Parson, Clifton, Bristol; T. Wormald, Oldham; B. L. Harding, Manchester; W. H. Ridge, Stafford. The election of members will be by ballot, at the Council meeting following the date of application, and the vote of two-thirds of those present, will be necessary to ensure election.

A book has been provided for the entry of names of members attending the meetings, and your Council have further arranged for the admission of visitors, each member having the privilege to admit two visitors, whose names and addresses should be entered in a column provided for the purpose.

The Council have nominated Mr Major Stewart of Liverpool, as first Vice-President of the branch, and they recommend that the Annual Meeting in April next be held in Liverpool.

An analysis of the Dentists Register shews that 1,400 names have been enrolled, of gentlemen engaged in the practice of Dentistry, in the Midland Counties.

Of these, about half are registered as practising in conjunction with pharmacy, and of these last, a considerable number—are names not to be found in the Chemists' and Druggists' Directory. Some difficulty is encountered in attempts to verify cases of incorrect registration on account of changes of residences, alterations or errors in spelling, &c., this pertains to both the Dentists' and Chemists' Directory, so that it is not easy to identify positively in all instances. If gentlemen, who may be aware of the facts in any locality, would be at the pains to collect them, and submit them in a concise form, so as to furnish clear and trustworthy evidence of

fraudulent registration, it would be of great service. Any one practising Dentistry, but who is not on the register, or anyone on the register, who can be shown not to have been in the practice of dentistry at the date of the Act, these are the two kinds of error it is desirable to expose.

It need scarcely be stated that an increase of members, both of the Central Association, and its separate branches, is absolutely necessary to enable the Executive to carry out the provisions of the Act, and there is little doubt that with moderate effort, we may all succeed in inducing others to join. The Association is destined to represent the profession in the future, and it is our business to make its representative character a living reality, and not a mere empty name.

Mr. J. S. CRAPPER (Hanley), moved the adoption of the report. It was seconded by Mr. Headridge (Manchester), and carried.

Mr. Major STEWART, of Liverpool, said that as his name had been mentioned in the report, he begged to thank them for the honour they had conferred on him by electing him Vice-President of the branch. He thanked them, on behalf of his brethren in Liverpool, for the honour the Society had conferred on them.

THE PRESIDENT'S ADDRESS.

The PRESIDENT then delivered his address, and said:—

Gentlemen,—It is my pleasing duty in the first place to thank you, not only for the kindness you have shown in electing me to the responsible position of President of your Branch, but also for having given me the honour of being the First President of the First Branch of the British Dental Association.

I need scarcely say that the fact of our meeting here to-day as we are now doing is a subject for congratulation, and the presence of so many, and the distance some of you have come, proves the interest you take in the great movement of the day, which has called into existence the central Society of which we now form a branch.

Every new movement is sure to give rise to more or less excitement and sometimes to great expectations, and I must, therefore, beg, first for myself, your kind forbearance for my numerous shortcomings, of which I am only too conscious; and secondly, for your society I must also ask your kind consideration, if at the first it seems to you scarcely to come up to the high standard of your anticipations, requesting you to remember that it is still in its infancy, that it has commenced its existence under difficulties, and I regret to say, not without some opposition. It will, therefore, require care and energy, not only on the part of the Executive body, but also in each individual member, to bring it to that state of maturity we all are anxious to see it attain. I feel certain from your presence here to-day that you have the interest of the Society at heart, but I venture to remind you that the simply feeling an interest in a thing is not sufficient in itself to command success. *That* is a result which requires, especially in these days, both energy and labour, and we must never forget that our future position and progress as a Society, will mainly depend on our own individual efforts, since we are each of us an individual unit, the aggregate of which units forms the Society itself.

It would be natural for you to expect in this, the first introductory paper, a history of the origin and objects of the Society, but these were so ably treated of by the worthy Secretary of the Society on his visit to Manchester,

and so many of you were present at the London meetings, and so much has been already written on the subject, that it would not only be superfluous but also presumptuous for me to attempt what has already been done by others so much more competent than myself, and I am therefore deprived of what would have been an interesting topic for an address; but although the past and the present have thus been taken from me, there still remains the future; and as I hope that in that future our Midland branch will ultimately form a not unimportant unit, I will therefore restrict what I have to say mainly to the subject of the branch itself, and in doing so must crave your kind indulgence for the many imperfections in my efforts, urging as my excuse the little time that numerous professional and other engagements leave at my disposal.

Here, I think, we may congratulate ourselves, that we exist as a branch of the British Dental Association and not as an independent Society, for I am sure we all feel that the scheme for the elevation of our profession to its proper position, which that Society has been formed to promote, has been most judiciously planned, and has thus far been most efficiently carried out—a scheme, which whilst it in no way discourages the attainment of the higher qualification of full membership of the Royal College of Surgeons, in those who are anxious to distinguish themselves, and possess the means and opportunity for so doing, provides an efficient education in all those scientific and practical subjects which it is necessary for a Dentist to know, and by means of the Dental Diploma, affords the public a guarantee that the possessor of that diploma has passed through a curriculum of such a standard as fully to qualify him for the practice of the branch of surgery which he professes. Nor must we allow to pass unnoticed the utter unselfishness of its promoters, who laboured out of pure love for their profession, knowing that they could never expect to see the good result of their labours in their lifetime, unlike the man who is said to have remarked that he never could see the good posterity had done for him that he need trouble himself about posterity: and as was the case with many of those grand memorials of the loving zeal and devotion of our ancestors, those noble cathedrals with which so many parts of our country are adorned, one generation was satisfied if they were able to lay the foundations and were permitted to see some portion of the superstructure erected during their lifetime, so in like manner, we must wait and not be discouraged if we are unable to see the full development of the present movement, but should

consider it a privilege to be permitted to assist in however humble a degree, in taking part in what we fondly and reasonably hope will prove the formation of a grand future for our branch of surgery.

To our Secretary and Treasurer are due, I believe, the credit of having originated the branch, and I take this opportunity of saying that I am unable in any way to claim a portion of that credit for myself, for I gather from a letter in one of the journals, that I was not only believed to have been one of the promoters, but that I was thought to have been guilty of an act of discourtesy in what I had done, or rather in what I had not done. I knew nothing of the project till I received a printed circular inviting my co-operation, and it was not till after the lapse of many weeks that I found myself able to join in the scheme. The promoters worked hard in drafting the bye-laws and arranging the other necessary preliminaries, but I am sure I may say for them that they feel amply rewarded by seeing their pet child, over which they have laboured so long, fully recognised as a legitimate offspring of the British Dental Association.

The objects of the branch, you see by the bye-laws, are fourfold:—

1. To render assistance, as far as possible, in carrying out the provisions of the Dentists Act.
2. The general consideration of subjects affecting the interests of the profession.
3. The reading and discussion of papers on Dental Surgery and Mechanics.
4. The cultivation of a generous professional spirit amongst practitioners throughout the district.

With respect to the first object, I think the purport of the Act is so clear that it requires but few remarks from me. It is impossible to draw any definite line where the help of an unqualified person must cease and that of the duly qualified practitioner commence. Any one who dresses a wound or a bruise, or reduces a dislocation, practises surgery, and no one could for a moment imagine that a law would ever be passed to prevent his doing so; and in like manner, we could not expect that the legislature would ever allow the drawing of a tooth or any other similar operation to be made a penal action, although the person so doing may be said to be practising Dentistry; but as soon as any one endeavours to make the public believe he is a qualified practitioner, by assuming the title of Dentist or any other name

implying the possession of the Dental diploma, so soon he becomes amenable to the law; the act thus guaranteeing to the public, that *for the future*, any one who claims the professional title shall of necessity have obtained the necessary qualification. Any person, or any number of persons, who may wish to put the act in force in any particular case can do so, but before taking action it will be necessary to obtain the sanction of the Medical Council. This at first sight may seem to be an unnecessary precaution, yet I think on further consideration you will admit that it is a very wise provision, as it entirely prevents the possibility of any one being proceeded against from personal pique or any other improper motive; and if in any case it is thought desirable that a person's name should be removed from the register, all that is necessary is to collect sufficient *reliable* evidence, and transmit it to the central board in London, who will bring the matter before the Medical Council, the only body by whom such action can be taken. And in cases where these proceedings may be necessary, I am sure they can be carried out without earning for ourselves the opprobrium of acting as spies or professional police, which some have already been willing to assign to us.

With regard to the second object of the Society, we must all feel that the interests of the profession will at times require the careful consideration of its members, and necessitate the existence of some organised body which shall be able to act with the authority of the bulk of its members, and for this, no better scheme can be devised than the one we are now so much interested in—the formation of a central society, with recognised branches in the more distant parts of the country, constituting an organisation by which the feeling of the majority of the profession may at any time be ascertained on any question that may arise bearing on the well being of the profession.

In the third object, “The reading and discussion of papers on Dental Surgery and Mechanics,” the surgery you will notice is placed first, and justly so, as the higher branch; though in early times, and I fear even to a more recent date, the order in importance was more frequently reversed in practice. The mere mechanical calling of former times—for in its infancy Dentistry was little else—has now been developed into a profession and gained admission within the sacred portals of the College in Lincoln's Inn Fields, and it is for the present and future generations to prove by the

exercise of their highest mental as well as mechanical faculties, that the profession is worthy of the position which has been accorded to it.

It has been noticed by those who most frequently attend the meetings of our speciality, that papers on mechanical subjects are more easily procured, and often prove more attractive than those on surgical subjects, but this will no doubt become less as the educational facilities of the present day are brought to bear more and more on the whole body of the profession. Surely the preservation of the natural organs is of far more importance and value to the patient than the substitution of others, however efficiently supplied.

What should we think of the surgeon who allowed himself to be deterred from directing all his energies to the preservation and restoration to health, of a diseased or injured limb, by the thought that an artificial substitute could be provided for it. No! No! Whatever our politics may be let our surgery be conservative. Far be it from me to appear to undervalue any branch of my profession, for no one can have been long in practice without having experienced the well earned gratification derived from noticing the relief from pain, and in many instances the perfect restoration to health, that follows the substitution of efficient members in the place of useless and diseased ones; but far greater is the credit and higher the appreciation of the patient, when the diseased natural organs themselves can be preserved and restored to a state of efficiency; and how great are the facilities for so doing in the present day, compared with the early reminiscences of many of our older brethren. All of us who were fortunate enough to hear the interesting paper read before the General Meeting of the Society in August last, must have been forcibly struck with the contrast between the paucity and quality of the instruments there mentioned, and the appliances of the present day. What would have been the feelings—I might almost say the bewilderment of the practitioner therein described, could he have been transported into one of the large dépôts with which we are so familiar. The numerous and beautifully adapted instruments for the variety of operations unknown in those days. The admirably adjusted forces for each form of tooth, the endless variety of excavators and pluggers, the wonderfully delicate nerve extractors, the rubber dam and its adjustments, the saliva pump, the electric mallet, the improvement in our chairs, and that greatest of all boons, both to patient and operator, the Morrison engine, the name of the inventor of which valuable instrument, ought to be indelibly inscribed in letters of gold in the Archives of Dentistry, to say nothing of the application of vulcanite

and celluloid, and the many ingenious appliances for the workshop. Surely these should lead us to value the benefits we enjoy, and teach us to strive to use them to the best of our ability, not influenced by the thought of self-glorification in attempting to surpass all others, in the performance of this or that brilliant operation, but ever remembering that the ultimate aim of all our efforts should be the increased amount of good which we are thereby enabled to accomplish for the benefit of our suffering fellow-creatures.

Although the *Transactions* of the Odontological Society contain a very voluminous and valuable collection of papers bearing on one speciality, there still remain numerous subjects and modes of operating which may be made productive of profitable discussion. Such are the replantation of teeth now attracting so much attention, the erosion of the surfaces of the teeth of which so little is known in the present day. The various improvements in the materials for filling, and amongst a variety of subjects, far too numerous to be mentioned here, the startling announcement of the so-called New Departure Creed.

As this last is a subject which I think we all feel greatly interested in, and also bears on the branch of Dentistry which we are now considering, I will venture a few remarks on some of the articles of the accepted and new departure creeds as tabulated in the *Dental Cosmos*.

I am not aware that the so-called accepted creed has been the recognised standard of practice in this country. The doctrine that gold, and nothing but gold, should be used for permanent fillings, has certainly been extensively promulgated by those of our Transatlantic brethren who have settled in this country; and I cannot but look upon this new departure, as the natural reaction which might be expected to follow the over anxiety to build up large adhesive gold fillings, on fragments of weak and often disorganized teeth, totally unsuited for such an operation; but until I can see some stronger reason than has been hitherto adduced by the advocates of this new departure, I must, in any case suitable for a good gold filling, confess my unwillingness to abandon for any other of the fillings now in use, a material that we know from past experience is capable, when judiciously applied, of preserving and restoring to a state of efficiency, in some cases for a period of twenty years or even for a much longer time, teeth which otherwise would have been lost in about the same number of months. For the efficient use of this material much must of course depend on the manipulative ability

of the operator, but it has often seemed to me a matter of doubt, whether, in the case of those large adhesive gold fillings, the patient has received an equivalent for the tedious and necessarily expensive operation that has been undergone.

The choice between contour fillings and separation of the teeth in the case of approximal cavities, must, I think, depend in a measure on the aptitude and judgment of the operator, since the two plans when successfully performed, may be made equally efficient. The latter however—separation—seems to afford greater facilities for operating, and also for cleanliness in those patients who are unable or unwilling to devote the time and attention necessary for that object.

The professed incompatibility of gold as a filling material with tooth bone, seems to me to be at variance with the frequent success which we all must have experienced, in the use of that filling in cavities on the labial surface of the roots of the upper incisors. I mention this particular position as being easy of access for operating, and also as being one where the enamel, tooth bone, gold, and saliva, are in constant contact with each other. In those cases where the success of the operation has not equalled our expectations, in endeavouring to estimate the probable incompatibility of the filling material as a cause of failure, it is, I think, necessary to consider whether in the ordinary operation for filling, the whole of the diseased dentine has been so thoroughly removed, as to admit of its proving a satisfactory test case, for the microscope reveals to us a change in the dentine in the sides of a decayed cavity, even when it appears sound to the eye, and also to the feel of the instrument; and in those portions of the margin of the cavity which we undercut for the retention of the filling, we have the dentine not only denuded of its nutrient covering, the periosteum, but also deprived of its nutrition from the pulp by the severance of the tubes of the dentine, and the intervention of the filling material, and in this deteriorated condition, in all probability still exposed to the continuous action of the same deleterious influences (whatever they may have been) which first caused the decay; and I cannot but think that the failure when it occurs, is in a greater measure due to the above named causes, rather than to the incompatibility of the gold with the tooth bone; but with all the success that has attended the use of gold for so many years, we must still, I am sure, be willing to admit the want of a perfect plastic filling, which shall

be able to withstand the friction of mastication, and also the deleterious action of the fluids of the mouth.

The dictum that, “a tooth that can be so treated as to be satisfactorily filled with anything, is worth filling,” is one, I think, that all who value conservative surgery must readily agree with.

That “unskilful and unscrupulous Dentists fill with tin covered with gold, thereby causing galvanic action, pulpitis, death of the pulp, abscess, and loss of the tooth,” has certainly not been an accepted creed in this country, for I was early taught the use of this material both alone, and in combination with gold, when I first commenced the study of my profession with Mr. Sheffield, of Exeter, and long experience, and the retention of a tin filling in my own mouth, in a perfect condition for over five-and-twenty years, has fully proved to me the value of this metal. When used in combination with gold, and exposed to contact with the fluids of the mouth, it certainly undergoes a chemical change, becoming nearly black in colour, but without staining the tooth as some amalgams do; it also becomes harder, and cuts harsh like an amalgam filling, but the change does not appear to cause any alteration in its bulk, or in any respect to interfere with its efficiency as a filling, nor have I ever found it produce any of the evils suggested in the creed, and I cannot but think that it would be more used, were it not for the universal prejudice that exists in the mind of the public in favour of gold, partly arising from the magical charm which resides in the word gold, and partly from the nothing but gold theory (if I may so call it), which has been so largely spread by our American brethren.

That “a filling may be the best known for a tooth and yet leak badly,” seems a simple admission that in some cases bad is the best that we can do for them. If experience has taught us anything, it certainly has proved that it is the leak which does the mischief; and as in warfare no fortress is considered stronger than its weakest part, so is it with a tooth that is filled. A small leaking point will soon undermine the filling, and prove its ruin.

The statement that “gutta-percha *properly used* is the *most permanent* filling material we possess,” must make us wish to know the meaning of the words, “properly used,” for the rapidity with which it wears away, in any position in which it is subjected to friction in mastication, seems to me to prevent the possibility of its ever being permanent.

The force of the article which says that “a poor gutta-percha filling, *in its proper place*, is better than a *good* gold one,” seems to depend on the question as to what is the proper place for a gutta-percha filling: surely not the place where we can insert a good gold plug, if by “good,” is meant one that is not only solid, but also sufficiently tight to exclude all moisture from the cavity.

It certainly is rather startling to hear from the country from which have come the severest criticisms on the use and the users of amalgam fillings, the admission that “amalgam per se, is an excellent filling material.” With amalgams, as with the other plastic materials, a *perfect* filling of its kind has yet to be discovered, for a measure of uncertainty seems to exist in all of them; but as a proof that an amalgam *may* make a good useful filling, I can say that I have seen more than one apparently in a perfect condition after thirty years’ wear, and I cannot but think that much of the discredit attaching to its use, arises from its being the filling we naturally have recourse to in all cases of doubt and difficulty.

That “the use of plastic filling material tends to lower the standard of Dentistry, thereby diminishing its sphere of usefulness,” is scarcely a fair way of stating the matter. The indiscriminate use of them would certainly do so, but the judicious selection of them in cases unsuited for the use of gold, need not necessarily interfere with the acquirement of the manipulative ability necessary for making good gold fillings, nor is it desirable that it should do so, as the power of manipulation required for the successful use of gold, must tend to perfect the powers of the operator in the use of all plastic materials, and must thus extend the sphere of usefulness of that Dentistry which has for its standard of excellency, ability to save teeth.

Time has only permitted me very briefly to notice some portions of this new creed, to show that it contains much that might be productive of profitable discussion, but I think from what has been said, we may reasonably conclude that each material has its own particular advantages, and that until we are in possession of a filling suitable for universal application, a judicious selection is necessary on the part of the operator in each particular case.

The fourth, and not the least important object of the society is “the cultivation of a generous professional spirit amongst practitioners throughout the district,” and to this I hope may be added throughout the

whole body of the profession; for the knowledge that we are branches of one central society, should lead to a feeling of fellowship not only with the members of the parent society, but also with the members of the other branches, and surely it is not too much to hope, that with the extension of the Society and its branches, and the feeling of professional brotherhood thereby engendered, we may see the gradual increase of that Christian charity which “thinketh no evil,” and which would scorn all attempts at self-exaltation, by the disparagement of the efforts and qualifications of a fellow practitioner, which I fear has been greatly encouraged by the isolation that has so long existed in our speciality.

Nor must we omit to notice the great benefit to be derived from the friendly discussion of subjects in which we are all mutually interested, not only to the profession at large, but also to each one individually, for which of us in his early days has not felt the wish that he had thought of this or that other mode of treatment in some particular case; and what more likely to impress new ideas on the memory, or, to use a familiar expression, make us “have them at our fingers’ ends” (the place where surely they are wanted when operating), than the friendly discussion of them which it is one of the objects of our Society to encourage?

The President of the Western branch has held out to us the hand of fellowship, which I am sure I need not ask your permission to be allowed to reciprocate most fully—not only in the wish for their future prosperity, but also in the assurance that any of their members who may have an opportunity of attending our meetings, will at all times receive that cordial welcome which it should be our greatest pleasure to extend to all the members of our Society and its branches, thus realizing one of the great objects of our Association, and proving not only by our words, but also by our actions, that we are anxious to encourage that professional intercourse which must tend not only to our mutual benefit, but also to that of the community at large; and if, as their President remarked, it is a pleasure to them to think that in the formation of our branch we show “the benefits resulting from the good example they have set us,” it is no less a subject of congratulation to us to know that they have at last recognised the good which is likely to accrue, both to themselves and to the profession generally, by following our example, and forming themselves into a local branch of the British Dental Association.

And now, fearing that I have already trespassed too long on your time and forbearance, it only remains for me to thank you for the kind reception you have accorded me, and to express the belief that, although the barque which we have so successfully launched will probably meet with the usual amount of storms and difficulties, still the old proverbial “long pull and a strong pull and a pull all together,” in which you are so able and willing to join, will carry it securely through the many undercurrents and quicksands which may threaten it on its course, and secure what we all so ardently hope for—a useful, as well as a long and prosperous career, for the Midland Branch of the British Dental Association.

Mr. MAHONIE (Sheffield), moved a vote of thanks to the President for his able and eloquent address, and in doing so, said that when the Association was weak, Mr. Campion had come to their aid in a very courteous manner, and they were now asked to give him their heartiest thanks for the paper which they had just heard read. The paper was one which would last more than a year. It was very instructive, and he (Mr. Mahonie) thought that Mr. Campion was eminently capable of giving them instructions in the way of Dental Reform. He moved “That the best thanks of this meeting be given to the President of the Midland Counties Branch of the British Dental Association, for the able and eloquent paper which he had just read.”

Mr. W. H. Nicol (Leeds), seconded the motion in suitable terms.

Dr. W. H. Waite in supporting the motion, said that it was a very great privilege for them to have such a President as they had in Mr. Campion. His wisdom had been of great service in superintending the efforts which the Executive have had to put forth. They had been delighted that morning with the interesting and instructive address which Mr. Campion had given them, but most of all with the very high and generous tone that breathed throughout the whole of the address, and he (Dr. Waite) thought Mr. Campion had struck a key-note which, when the address came to be printed, would vibrate not only throughout the Midland Counties of England, but through every country where the Dental profession was carried on. The motion was carried with much enthusiasm, and the meeting then adjourned till the afternoon.

The Members assembled in the Memorial Hall at 2.30 p.m., Mr. Campion in the chair.

The PRESIDENT stated that some idea had been started of bringing specimens to the meeting, and in order to show what curiosities might be casually collected, he offered a few specimens to the notice of the Members.

Mr. ROFF KING, Shrewsbury, then read a paper on “Modelling and Modelling Composition.”

Mr. J. S. CRAPPER (Hanley) read a paper, entitled “New Ideas in Mechanical Dentistry,” and exhibited specimens of teeth which had been manufactured by a new process and sent to him to exhibit.

The SECRETARY read a paper received from Mr. F. Richardson, Derby —“Has Vaccination any Effect on the Degeneration of the Teeth?”

Mr. H. MARSH (Manchester) read a paper on the “Use of Nitrous Oxide Gas.”

Mr. MURPHY desired to ask the Chairman whether Licentiates in Dental Surgery were legally responsible if a patient expired whilst under the influence of the nitrous oxide gas, or subsequently to the administration.

In the discussion which followed, there appeared to be a unanimous opinion amongst those present that Licentiates were not responsible, but that it would be impossible to define such a position.

The SECRETARY read a letter from Mr. Kyan of Preston, in which that gentleman expressed his regret at his inability to be present, and hoped the Members would have a successful meeting.

Mr. RENSHAW moved a vote of thanks to those gentlemen who had furnished them with such interesting and instructive papers.

Dr. D. A. WORMALD, in seconding the resolution, said they could, with a little energy and support, make that Society very successful, and as a representative body on behalf of their profession in the Midland Counties, they could maintain such a position that it would become an honour to belong to it.

The CHAIRMAN remarked that it was no light work to prepare such papers as they had heard read; much time and trouble must have been spent in the digest of the various subjects. He gave the hearty thanks of the meeting to the contributors of those papers.

The resolution was carried unanimously.

Mr. Major STEWART proposed a vote of thanks to Mr. Campion for presiding.

Mr. KING seconded the motion, which was carried unanimously.

The CHAIRMAN (Mr. Campion) returned his best thanks for the kind reception and assistance the Members had rendered him whilst in the chair.

THE DINNER.

In the evening a dinner was held at the Queen's Hotel, Mr. Campion presiding, Mr. Major Stewart in the vice-chair.

The CHAIRMAN proposed the health of "The Queen and the rest of the Royal Family."

Mr. MAHONIE then gave a recitation from Shakespeare.

Mr. Major STEWART in rising to propose the toast of the President of the branch said—It is my pleasing duty as your Vice-President to propose this toast. I find our worthy Chairman took his M.R.C.S. degree in 1856, I therefore regard him as the father of the profession in this district of England; those who were present this morning heard his inaugural address. Those words were words of wisdom, and were addressed not only to the older, but also to the younger members of our Branch Association. His address will be read with pleasure by members of our profession in England and America. With regard to the chairmanship of this branch, I think it would have been wise if Mr. Campion had been elected for three years, until we were more consolidated. I am glad to see so many members from this district present this evening, and will now conclude by asking you to drink a bumper to the long life and successful career of our Chairman. The toast was accepted most loyally.

The CHAIRMAN in responding said, gentlemen, you must have listened to my voice sufficiently to-day—this is my last appearance before you this session. Our Secretary has promised to allow me to enjoy myself, *otium cum dignitate*, after I have returned my best thanks. It is with great diffidence I respond to this toast, I really feel the difficulty increased tenfold by the kind manner in which you, sir, have proposed, and my friends accepted this toast. One word in regard to sub-branches—I can see no reason why such large towns as Liverpool and Manchester should not establish sub-branches; frequent meetings, say once a month, could be held for an exchange of thoughts and methods of operation in our department of Surgery, in which we are all interested; so constituted, they would be private societies, to which any member of our branch might be admitted. The great event of the next year will be the meeting of the International

Medical Congress, held in London during August. This Congress has held its meetings every alternate year for the last twelve years. The London members of the Medical profession have taken it up very warmly, and are making preparations, which must result in giving the Congress such a reception as will be worthy of our capital.

The President of the Dental section is Mr. Edwin Saunders, whilst the Vice-Presidents are Mr. John Tomes, and Mr. Spence Bate, Mr. C. S. Tomes acts as Secretary, to all of whom we are much indebted for their untiring efforts in raising our department of Surgery to its present position. I trust that our branch will be well represented at this Congress.

Another subject I will broach is the establishment of a Dental School in Manchester. If such a school is formed it should be in concert with the Manchester Medical School at the New Victoria University. When the charter was first granted for the formation of a university, a medical school was not included, but there appears very little doubt that in the course of time such school will be established; let us then wait patiently until we can thoroughly and efficiently carry out such an undertaking.

I am sorry I cannot introduce you to a Dental hospital; this is to be deplored, more especially as we took a prominent part in Dental reform for the good of Dental practitioners. If younger and more able men will commence the work, they shall receive my heartiest support and co-operation. I must now conclude by expressing regret that I have not the gift of speech, that I might explain to you the warm feelings I have in regard to the reception you have given me, and the kind manner in which you have drunk my health.

Dr. D. A. WORMALD sang "The Spinning Wheel."

Dr. WORMALD, in proposing "The British Dental Association," said: Mr. President, and gentlemen, this toast is one, I am sure, you will drink with the utmost cordiality. Those who sit round this board, and have taken a part during the past few years in the politics of our profession, must experience a feeling of congratulation that we are able to meet to-night, and realize the fact that at last we occupy a recognized position. We need re-organization, and we need a central body, and the British Dental Association is that recognized body, and demands our support and assistance. Well, join that Association, and do what you can to lift up the profession from the low level in which we have been placed, and raise it to a much higher level. We

need, as your President so kindly put it this afternoon, a higher professional tone, and a more generous feeling towards our brethren; and that can only be obtained by coming more and more in contact with each other, and through the influence of such an organization as this we are met to commence, which will assist in placing our profession in a position which it should long ago have occupied. Many of those who are now in our ranks may not see the full result of our labours, but depend upon this, that our labours will be recognised in the great difference which will show itself more and more in this country between those who are worthily following their calling, and those who indulge in non-professional practices. The British Dental Association will become strong and durable, and then we may be sure that under the wisdom and the guidance of those who have piloted our steps so far, we may safely follow them wherever they lead, and so lay a foundation upon which the rising generation may build up their professional career, and help to raise the status of our profession. That profession is worthy of all the time and attention that any honourable gentleman will give it, if he will follow his calling worthily, and serving the interests of a suffering public. In connection with this toast, Mr. Chairman, my friend Dr. Waite, will have the pleasure of responding, and I have the greatest pleasure in coupling his name with it.

Dr. WAITE responding, said, Mr. President, Mr. Vice-President, and gentlemen, it is to me both a pleasure and a pride to be permitted to respond to the toast of the "British Dental Association." I am not much of a believer in post-prandial oratory. The period of replenishment for the physical energies, should be a period of repose for the energies intellectual, or perhaps better still, a time of refreshment, by agreeable and amusing conversation. That Englishmen should be fond of dining together, is natural and characteristic; that they should be almost equally prone to after dinner speechifying is somewhat surprising. It is well, however, when the usages imposed by custom can be observed, without making a too serious demand upon our drowsy powers, and fortunately we have a condition of things existing at the present time which, when fairly estimated, is calculated to awaken agreeable reflections, and inspire us with salutary hopes.

Within the short space of five years, events have occurred with astonishing rapidity, that have substantially and permanently altered the position of the Dental profession, not alone in its relation to the general public, but also in its internal organisation, and the mutual relation of

individuals to the whole body. These events may be classified in three distinct items, and in each we shall find elements of progress largely preponderating. First of all, most of you will recall the feeling which prevailed when the idea of obtaining an Act of Parliament was invoked at the Manchester meeting of 1875—it was not regarded exactly as an utopian idea, but certainly we felt it to be something almost too good to hope for. Well, in less than three years from that time the Dentists Act was passed, and now, after two years experience and consideration, we are able to discover what it is we have obtained, though it is all too soon as yet, to realise (except very partially) the practical benefits the Act is destined to confer. This much, however, we know. We have a legal status. We have legal rights and privileges. We are admitted to the fellowship of the honourable and learned professions. These three points are assured, and we should see to it, that we interpose no barrier of professional character or conduct between ourselves and the privileges our Act bestows. We know, moreover, that there are secured for those who will come after us, more substantial benefits of education, of practice, of public appreciation, such as none of us have been permitted to enjoy. This is the characteristic feature of the Dentists Act. Its noblest provisions are for posterity. No change, nor circumstance, can deprive any who have assisted in passing this Act, of the pure satisfaction of knowing that they have helped to hand on the profession of Dental Surgery in a far better condition than that in which they received it. Many have contributed in various ways, whose share in the work will be forgotten, but the work remains. Identified with the whole of the movement, the names of Tomes and Turner will be preserved and revered for many, many years, yet withal the abundant reasons for personal gratification which those gentlemen have, I am sure they rejoice a thousand fold more over the results achieved, than they do at any honour or advantage accruing to themselves. Take it as it stands, acknowledging its few imperfections, and recognising its many decided benefits, we have as good an Act as we had any right to expect, and as good as it was possible for us to get.

One of the hindrances to Dental advancement has hitherto been the scarcity of educational facilities, and the small proportion of practitioners who possessed any Dental qualification. At the time to which reference has already been made, there were only some 300 gentlemen in the whole of Great Britain who held a Dental diploma. There was only one licensing

body in the country which granted Dental certificates. Now we have 630 gentlemen holding recognised Dental qualifications, viz., of London about 400, Edinburgh 16, Dublin 180, Glasgow 33, and the three last mentioned bodies are offering to gentlemen in practice, the opportunity of examination, *sine curriculo*, upon thoroughly reasonable and accessible terms. Not only so, the Colleges of Ireland and Scotland have caused their certificates to possess a peculiar value, in the careful provision they have made, for preventing unprofessional practices on the part of their licentiates.

I am certainly astonished that the number of practitioners who have already availed themselves of these facilities is so small. I know that this matter of obtaining a Qualification is regarded variously by different persons, but nothing is more certain than the growing tendency toward accredited qualification as indispensable to professional status; the public are rapidly becoming aware (and it is our business to educate the public in this matter) that the operations of Dental Surgery cannot be safely entrusted to uneducated heads or hands, and as a guarantee of something like the necessary knowledge and skill, the public are beginning to understand the value of a purely Dental Qualification. The larger the number of Licentiates the more quickly the public will be taught, and ere long the unqualified Dentist will become a mere historic fact.

The all important feature of recent progress is the British Dental Association. The Dental body has been afflicted with inherent weakness through the want of organisation, a weakness that paralyses, and at the same time, a weakness endowed with remarkable power, the power of successfully defeating efforts toward general reformation. So long as we were isolated and detached it is not wonderful that we were selfish and afraid of one another, but a new era has dawned. We have an Association to which all are eligible, where all can meet on the common ground of professional brotherhood and intercourse. The control of the Association, embracing the future welfare of the whole body, is committed to a Board constituted on a representative basis. None need be excluded from participation, save by their own poverty of professional spirit, a poverty exhibited alike by persistent recourse to unprofessional practice, and by selfish indifference to professional politics. There may be a kind of superiority in having a large practice, or in holding a high scientific position, but unless I am greatly mistaken the true superiority is that which

overleaps all selfish considerations, and delights in consecrating whatever power or position we may have to the elevation and improvement of the whole profession. Thank God we have men who have been inspired with a large measure of true professional spirit. Men who could think for themselves and hold their own opinions, but who at the same time have been large-hearted enough to merge minor differences in the pursuit of one grand object. Such men as Edwin Saunders, John Tomes, J. Smith-Turner, T. A. Rogers, A. Coleman, Dr. Walker, our esteemed President, and many others, stand forth as examples of *true* as distinct from *sham* superiority.

The facilities now afforded by our separate branches have removed the excuse so often urged when all the meetings took place in London, and there is practically no obstacle to our speedy improvement. We possess advantages of a political nature quite equal to those existing in any kindred profession.

These are the bare facts of the present time; they furnish material for much thankfulness and for cheerful anticipation. They are the foundation on which our future must be erected. What that future is to be depends on what we mean to be. The scaffolding is up, the materials are ready. Willing hearts and willing hands are needed to complete the edifice. There is a sublime principle working through the whole human family, the operation of which no individual can withstand. It was formulated by the Founder of the grandest system the world has ever known, and this is the formula—

“Whosoever would be chief among you, let him be your servant.”

Mr. MAHONIE, in proposing the toast of “Sir John Lubbock, and the promoters of the Dentists Act,” said:—No words of mine can add additional lustre to the names—Sir John Lubbock, Mr. John Tomes, Mr. James Smith Turner. These men have earned a name for themselves which neither crayons nor oils—not even if they were placed in cathedral aisles—could add more; these men have laboured much—laboured hard; I beg to propose their health.

Captain ROGERS, in responding, spoke of the great triumph Sir John Lubbock had accomplished in passing Bills connected with science, this Parliament. He had great pleasure in thanking them for the hearty manner in which they had received the names of these gentlemen connected with Dental Reform.

Mr T MURPHY in proposing the “Manchester School of Medicine” said:— I have had put into my hands a toast to propose, which is so intimately connected with our branch of the profession, that it will require no eulogy on my part for it to be received with acclamation, I mean the Manchester Royal School of Medicine. Perhaps it will not be out of place if I give you a short outline of its history. It was founded by the late Mr. T. Turner in the year 1824; it was known for many years as the Pine Street School. In 1850 a second school was established, and being connected as I then was with a hospital in this city, I can well remember the jealousy there was about the subjects that were sent out; first one school and then another complaining that they were not getting their fair share, and I must admit that the leaning was towards the old school, which I think got, on the whole, rather more than its share. In 1858 the founders of the rival schools came to the conclusion that the interests of the students would be best promoted by an amalgamation, forgetting their differences and working together for the common weal. In this way, from the Union of the Pine Street and Chatham Street Schools, that school arose which has been long known as the Manchester School of Medicine. For sixteen years the work was carried on in Faulkden Street with difficulty both with regard to accommodation and light. About the year 1872, a meeting of the Owen’s College authorities met the leading Members of the medical staff, and it was agreed that it would be for the welfare of the School if it were handed over to the Owen’s College authorities. The result has been the erection of one of the finest Medical Schools in the kingdom, replete with everything necessary for the study of medicine. The course of instruction not only enables students to qualify for College and Hall, but affords the more aspiring every facility for qualifying themselves for the higher distinctions of their profession. When I mention the names of Turner, Jordan, Wilson, Smith, Wilkinson, Bradley, Southam, and last, but not least, the celebrated Dalton, and others, who have gone; Gamgee, Watson, Williamson, Roscoe, Roberts, Morgan, Lund, Simpson, Ransome, and other able men, who are at present at the helm, can there be a doubt about the quality of the education given? and, in conclusion, is it too much to ask that before long we should have established in this city, and in connection with the Victoria University, a Dental School where we can send our sons to receive at the hands of these able men the necessary training to fit them for the practice of Dentistry.

Mr. MARSH returned thanks for the toast; he trusted some day Mr. Campion would be appointed Consulting Dental Surgeon to the Manchester School of Medicine.

Dr. WAITE proposed "Success to the Western Counties Association," coupling with it the name of Mr. J. C. Parson, of Clifton.

Mr. PARSON having responded, the proceedings terminated.

Correspondence.

We do not hold ourselves responsible for the views expressed by our Correspondents.

MR. SEWILL ON DENTAL CARIES.

TO THE EDITORS OF THE "MONTHLY REVIEW OF DENTAL SURGERY."

SIRS,—While thanking you for your courteous notice of my contribution to the second edition of Dr. Fothergill's work, permit me to point out that my contribution was not intended as an exhaustive discussion of the subject of dental caries. I was asked to write only upon the "Hygiene of the Teeth," and to indicate to the medical practitioner the means available for the prevention of dental disease. In accordance with the scheme of the work it was requisite to explain the etiology and pathology of caries in order to make clear the *rationale* of treatment, and therefore, with brevity consistent with the limited space at my disposal, I did so, adopting mainly what is termed the "chemical" theory of the disease. This view, as you remark, is certainly "theoretical," but I venture to say it approaches as nearly to demonstration as need be. I certainly do not hesitate to accept it as fact, and I felt more bound to adopt it in the present instance, seeing, at least, that upon this theory both the prophylaxis of caries as well as our treatment of the disease seem based.

I should be glad to see a discussion of the subject started in your columns or at the Odontological Society.

6, Wimpole Street,
September 30th, 1880.

Yours faithfully,
HENRY SEWILL.

Birmingham Dental Hospital, Broad Street.

Consulting Physician.—James Sawyer, M.D., Lond., M.R.C.P., Physician to the Queen's Hospital.

Consulting Surgeon.—James West, F.R.C.S., Senior Surgeon to the Queen's Hospital.

Consulting Dentists.—Thomas H. English, Adams Parkes, L.D.S.Eng.

Surgeon Chloroformist.—F. H. Maberley, M.R.C.S.

Dental Surgeons.—Charles Sims, L.D.S.Eng., Wednesdays; H. Breward Neale, L.D.S.I., Tuesdays and Thursdays; F. H. Batchelor, L.D.S.I., Mondays and Fridays; F. E. Huxley, M.R.C.S.Eng., L.D.S. Edin.; Saturdays.

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DENTAL HOSPITAL OF EXETER.

CASES TREATED FROM JUNE 14TH TO SEPTEMBER 30TH, 1880.

	{ Children under 14	309
Extractions.	{ Adults	641
	{ Under Nitrous Oxide and Ether	29
Gold Stoppings		45
White Foil ditto		37
Plastic ditto		127
Miscellaneous cases		129
	Total	1,317

HENRY BROWNE-MASON,
Hon. Sec.

White Nickel Bronze.

The pure white tint of nickel, its inoxidisability, and, therefore, the ease with which it is kept clean, have caused it to come into general use for metal fittings, especially those of vessels and carriages, as well as for the bright parts of engines and machines. But, on account of the relatively high price hitherto obtained by nickel, it has been employed chiefly in the state of electro-deposit.

Nickel-plating is effected by two processes, one hot and the other cold. In the former a solution of sulphate of nickel is used in an enamel cast-iron trough. This is the more rapid method, but its results are not so satisfactory as in the cold process. The latter is effected with a bath of double sulphate of nickel and ammonia, the articles to be nickelised being suspended by a platinum wire attached to the positive pole of the battery, while a plate of pure nickel, to restore the metal to the bath, is put in connection with the negative pole. The nickelisation thus produced, though it requires a longer time than the other process is finer and more durable.

There are, however, several objections to nickel-plating. The process is injurious to the workmen engaged in it. The thin coating of nickel is liable to peel off, and in the case of parts exposed to wear, or even with cleanings, in course of time, the nickel becomes so worn off at the edges as to expose the base metal beneath. Besides, nickelisation does not entirely prevent the oxidation of steel and iron.

A French Company, La Société Française Anonyme de Nickel, briefly referred to in connection with the Applied Science Exhibition, Paris, are endeavouring to supersede the nickelisation of brass and copper by the use of solid nickel bronze, which is thus obtained. The ore, *garnierite*, extensive deposits of which exist in New Caledonia, the French penal settlement, is worked by Messrs. John Higginson and Co., of Noumea, the

concessionaires. It is by them subjected to a roasting on preliminary fusion, which forms a regulus, containing from 60 to 70 per cent. of metallic nickel. This substance is shipped to the works of the company at Septemes, near Marseilles, where it is smelted into ingots and granule, containing 99½ per cent, of pure nickel, and ¼ per cent, of utilisable metallic substances. The extent of the mineral deposits in New Caledonia, the reduction in the cost of freight owing to the concentration of metal through the preliminary fusion, and the economy effected by the new methods of reduction devised by M. Jules Garnier, enable the company to sell the pure metal at about one-half the price it obtained three years ago.

For a long time past, efforts have been made to work nickel, but without success, until the present year, because it was found too brittle; now, however, pure nickel has been rolled and forged; cups, knives, and other articles have been formed of it. As a rule, the pure nickel is formed by the founder with various proportions of copper, zinc, and tin, according to circumstance, to form nickel bronze; but at least 20 per cent. of nickel is required to secure inoxidisability, and to give the desired tint. All articles that are now made of brass or copper nickelised may be produced in solid white nickel bronze, by the same processes, and with the same plant, and at practically the same cost; they are also 20 per cent. stronger, so that in many cases they may be made so much lighter. The great strength and the property of non-oxidisation render this alloy eminently suitable for mathematical and musical instruments. A small quantity of nickel added to steel increases its hardness, and renders it inoxidisable, while edge-tools made of the alloy stand better than those of ordinary steel. A nickel bell metal is also found to give good results. The Paris Exhibition of 1878 proved the action of a totally new system of metallurgy in connection with this beautiful metal, and that of 1879 showed its practical introduction into most branches of manufacturing industry.—*Journal of the Society of Arts*.

TO CORRESPONDENTS.

All communications intended for the Editors should be addressed to them at 22, Grosvenor Street, Grosvenor Square.

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1. Silently corrected typographical errors and variations in spelling.
2. Retained anachronistic, non-standard, and uncertain spellings as printed.
3. Footnotes have been re-indexed using numbers.

*** END OF THE PROJECT GUTENBERG EBOOK THE MONTHLY
REVIEW OF DENTAL SURGERY, NO. VIII. OCTOBER, 1880. VOL. I.

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